

GAURAV ALOK GUPTA

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EDUCATION

MSc., Computer Science (Average Grade: 16.45/20) - University of St Andrews, UK | Sept 2025 - Sept 2026 (Expected)

B.Tech., Computer Science and Technology (CGPA: 8.88/10) - Dayananda Sagar University, India | Aug 2019 - May 2023

PROFESSIONAL EXPERIENCE

TALLY SOLUTIONS PVT. LTD. - BENGALURU, INDIA | June 2023 - June 2025

Software Development Engineer II | C++, Python, AWS, OCI

- Engineered an **automated failover and self-healing system** in Tally Prime on AWS, restoring operations for **30+ client systems** during outages. Recognised with the **SPOT Award as Customer Delight Champion**.
- Developed **20+ custom Lambda functions** across **10+ components** enabling **zero-loss cloud platform migration** between AWS and OCI, eliminating vendor lock-in and improving scalability.
- Improved core product stability by resolving critical defects in **XML and Math Parsers** and the **Tally-to-Tally migration tool**, preventing data loss for enterprise customers.
- Delivered key product features - **file load source identification** and **placeholder collection attributes** - in TallyPrime Developer, strengthening third-party TDL integration.

PROJECTS

PRIMAROS: DRUG-DRUG INTERACTION PLATFORM | MSC DISSERTATION | Python, FastAPI, PostgreSQL, Redis

- Architected a **layered clinical decision-support system** integrating three heterogeneous medical data sources (DrugBank, OpenFDA, RxNorm) into a unified domain model of **21,500+ drugs** and **2.9 million directional interactions**, with streaming ingestion pipelines that process **1.8 GB DrugBank XML** and **8 GB OpenFDA label set** with flat, file-size-independent memory.
- Built a **REST API** with **trigram-indexed fuzzy drug search** and a **Redis pair cache** that cuts interaction-check latency, with schema migrations via **Alembic** and containerised deployment via **Docker Compose**.

INTERVIEW PREPARATION AGENT | Python, LangGraph, Gemini API, Tavily API | [GitHub](#)

- Built a stateful **LangGraph agent** with three modes: CS/networking revision mode **dynamically adapts question difficulty** using **per-topic confidence scores** persisted in SQLite, behavioural coaching uses **structured LLM extraction to score and rewrite answers** in STAR format, tracking competency coverage across sessions. Question banks self-update via **Tavily web search** with deduplication.
- Designed a **CI pipeline** where an **LLM-powered code reviewer runs on every push and PR**, posts inline review comments with severity-graded findings on pull requests, auto-fixes critical issues on pushes to main, with **retry/backoff for rate-limit resilience**. **115 unit tests with full LLM mocking** run on every commit.

SKILLS

LANGUAGES AND FRAMEWORKS | C++, Python, JavaScript, TypeScript, Haskell, Erlang, React, FastAPI, HTML, CSS, Sass

CLOUD AND DEVOPS | AWS (Lambda, EC2, S3), Oracle Cloud Infrastructure (OCI), Docker, CI/CD, GitHub Actions

AI AND INTELLIGENT SYSTEMS | Gemini API, LangChain/LangGraph, LLM Integration

DATABASE TECHNOLOGIES | SQL, PostgreSQL, MongoDB, Redis

TESTING AND TOOLS | pytest, Unit Testing, Git, GitHub, Jira, Confluence, Figma, Postman, Claude Code

EXTRACURRICULARS AND VOLUNTEERING

- Process, Methodology, and Tools (PMT) trainer and Hackathon Mentor for 10 teams (2024). Tally Solutions Pvt. Ltd.
- Shortlisted for JPMC Code for Good Hackathon (2022). JP Morgan Chase & Co.
- C Language Trainer for 70+ students and pre-placement session lead. DSU (University)